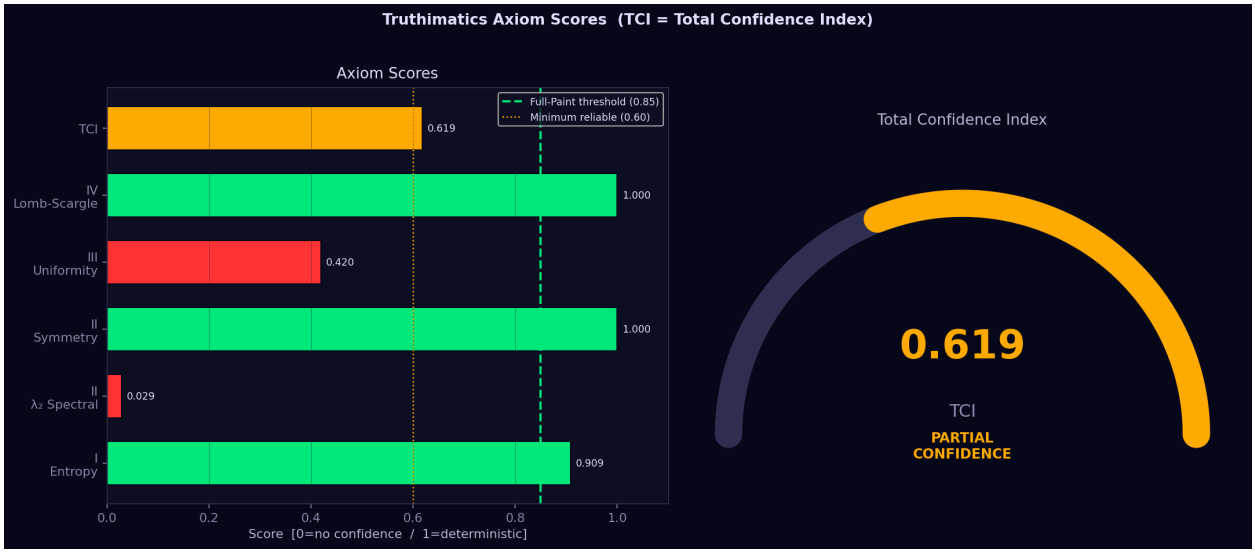
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_5E1412FCA7
Taxa	Pterodactylus
Clade	Pterosaur
PBDB Records	13
Mass (regression)	233.1 kg (Campione & Evans 2012)
Mass (volumetric)	56.3 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	0.072 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6185 (partial)
Axiom I Entropy	0.9092
Axiom II λ■	0.1171 (symmetry 100%)
Axiom III Uniformity	0.4199
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	66.7% Measured / 33.3% Inferred
Elements	24
CoG (m)	[0.077, 0.229, 0.0]
Generated	2026-05-01 00:04 UTC

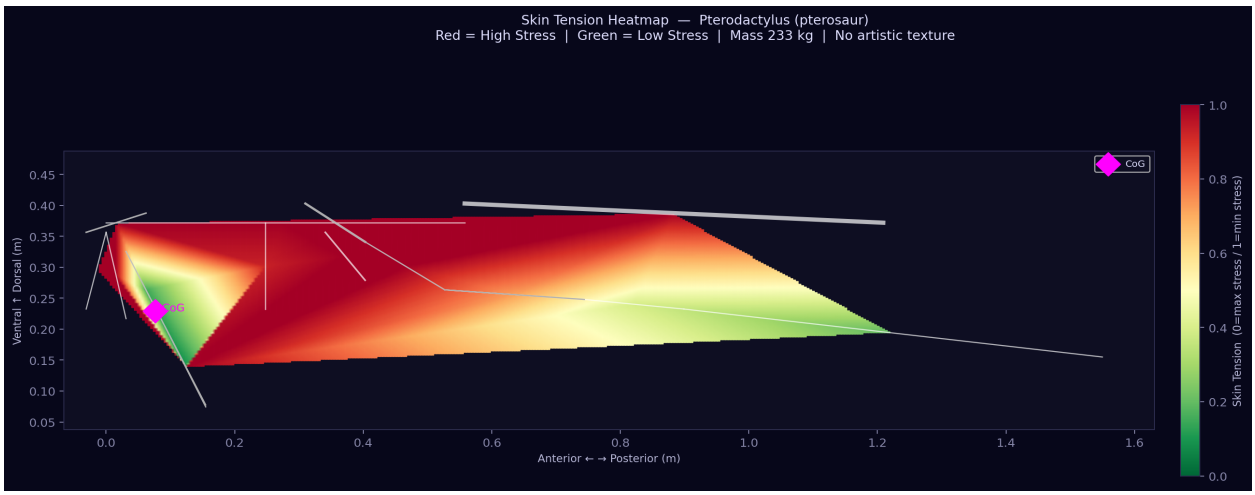
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



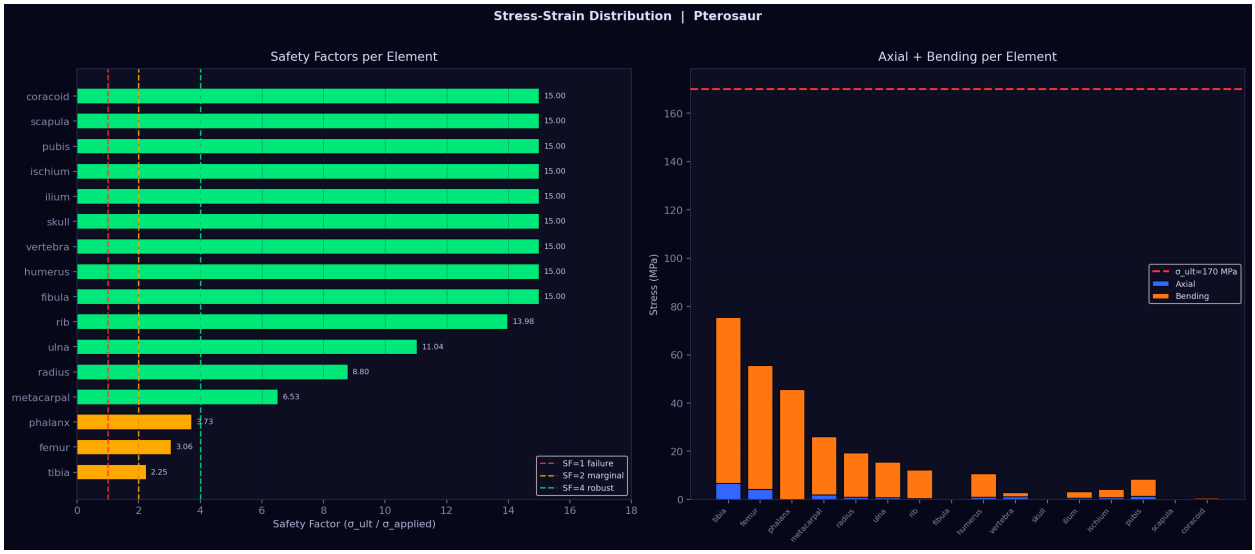
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

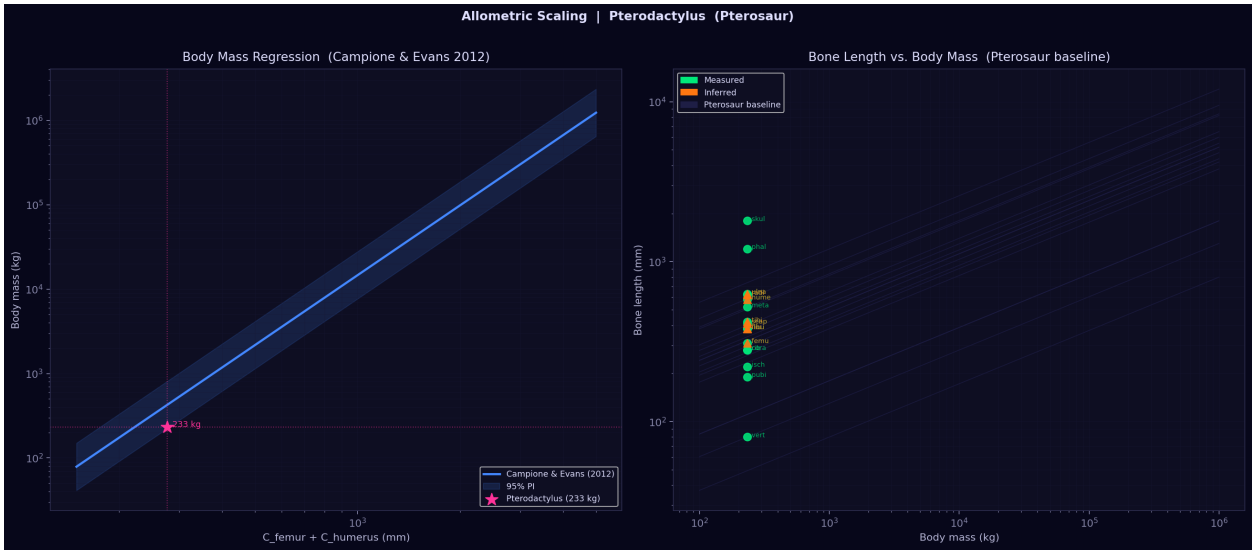


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.156	0.580	0.737	15.000	OK
femur	4.115	51.356	55.471	3.065	WARNING
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.896	9.765	10.661	15.000	OK
ilium	0.574	2.621	3.194	15.000	OK
ischium	0.706	3.500	4.206	15.000	OK
metacarpal	1.892	24.156	26.048	6.526	OK
phalanx	0.000	45.627	45.627	3.726	WARNING
pubis	1.255	7.166	8.420	15.000	OK
radius	0.922	18.406	19.328	8.796	OK
rib	0.373	11.785	12.159	13.982	OK
scapula	0.093	0.392	0.485	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	6.666	68.853	75.519	2.251	WARNING
ulna	0.782	14.614	15.396	11.042	OK
vertebra	1.067	1.749	2.815	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.839 m	Femur + tibia + metatarsal
Estimated Speed	5.5 km/h (1.52 m/s)	Gait: Walk | Fr=0.28
Stride Length	1.314 m	Alexander (1976) / Thulborn (1990)
Bite Force	1826.4 N (186.2 kgf)	Pterosaur regression (Witton 2010)
Skull Length	1800.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.2838	Circumference / length (dimensionless)
Limb Strength Index	1.3e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	2.79 m	Allometric from femur length
Est. Neck Length	0.84 m	Allometric from femur length
Est. Tail Length	0.22 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	280.0	85.0	810.32	1011.5	Measured	0.88	15.00
femur	310.0	28.0	97.35	121.5	Measured	0.88	3.06
femur_right	310.0	28.0	97.35	121.5	Symmetry	0.84	3.06
fibula	380.0	8.0	6.88	7.2	Measured	0.88	15.00
fibula_right	380.0	8.0	6.88	7.2	Symmetry	0.84	15.00
humerus	580.0	60.0	836.35	1044.0	Measured	0.88	15.00
humerus_right	580.0	60.0	836.35	1044.0	Symmetry	0.84	15.00
ilium	380.0	75.0	856.18	1068.8	Measured	0.88	15.00
ilium_right	380.0	75.0	856.18	1068.8	Symmetry	0.84	15.00
ischium	220.0	40.0	140.99	176.0	Measured	0.88	15.00
metacarpal	520.0	45.0	157.13	129.3	Measured	0.88	6.53
phalanx	1200.0	30.0	130.29	100.9	Measured	0.88	3.73
pubis	190.0	30.0	68.49	85.5	Measured	0.88	15.00
radius	620.0	35.0	304.22	379.8	Measured	0.88	8.80
radius_right	620.0	35.0	304.22	379.8	Symmetry	0.84	8.80
rib	280.0	12.0	16.15	20.2	Measured	0.88	13.98
scapula	410.0	110.0	1987.14	2480.6	Measured	0.88	15.00
scapula_right	410.0	110.0	1987.14	2480.6	Symmetry	0.84	15.00
skull	1800.0	220.0	34896.18	43560.9	Measured	0.88	15.00
tibia	420.0	22.0	81.42	101.6	Measured	0.88	2.25
tibia_right	420.0	22.0	81.42	101.6	Symmetry	0.84	2.25
ulna	630.0	38.0	364.39	454.9	Measured	0.88	11.04
ulna_right	630.0	38.0	364.39	454.9	Symmetry	0.84	11.04
vertebra	80.0	55.0	96.93	121.0	Measured	0.88	15.00